WORKED SOLUTION

STEP I Specimen paper, Question 14

Probability and Statistics • Expected Value

STEP I Specimen paper, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s1-q14

The key idea

Everything depends on one table: the distribution of the total of two fair four-sided dice. Once we have it, the expected gain for any guess x is just "prize times chance of winning, minus penalty times chance of losing". The interesting tension is that large guesses carry large prizes but small chances, so the best guess is not obvious until we compare them all.

Solution

Let S be the total score. Each die shows $1, 2, 3, 4$ with probability $\frac{1}{4}$ each, independently, so each of the 16 ordered pairs is equally likely. Counting the pairs with each total gives

s	2	3	4	5	6	7	8
$\mathrm{P}(S=s)$	$\frac{1}{16}$	$\frac{2}{16}$	$\frac{3}{16}$	$\frac{4}{16}$	$\frac{3}{16}$	$\frac{2}{16}$	$\frac{1}{16}$

(for example, $S=5$ comes from $(1,4),(2,3),(3,2),(4,1)$).

If B guesses x and $p=\mathrm{P}(S=x)$, his gain G in pence is $100x^2$ with probability p and $-50x$ with probability $1-p$, so

$$\begin{aligned}\mathrm{E}(G) &= 100x^2p - 50x(1-p)\\ &= 50x[(2x+1)p-1].\end{aligned}$$

Writing $p=\frac{k}{16}$, where k is the number of ways of making x,

$$\mathrm{E}(G)=\frac{25x[k(2x+1)-16]}{8}.$$

Part (i)

Only $(4,4)$ gives a total of 8, so $p = \frac{1}{16}$ and

$$\mathrm{E}(G) = 6400 \cdot \frac{1}{16} - 400 \cdot \frac{15}{16} = 400 - 375 = 25,$$

so B's expected gain when guessing 8 is 25 pence, as required.

Part (ii)

A guess outside $2, 3, \ldots, 8$ can never win, so (taking x positive, as a score must be) it only loses money. For the seven possible totals, the formula above gives

x	2	3	4	5	6	7	8
k	1	2	3	4	3	2	1
$\mathrm{E}(G)$ (pence)	-68.75	-18.75	137.5	437.5	431.25	306.25	25

For instance, for $x = 5$: $\mathrm{E}(G) = \frac{25 \cdot 5\,(4 \cdot 11 - 16)}{8} = \frac{125 \cdot 28}{8} = 437.5$.

Ask yourself The prize grows like x^2, so why isn't a big guess like 7 or 8 best?

The prize does grow quickly, but the chance of winning falls off on either side of 5, and the penalty is paid on the much more likely event of being wrong. At $x = 8$ the chance has dropped to $\frac{1}{16}$, which almost exactly cancels the large prize. The two effects balance best near the middle: $x = 6$ has a bigger prize than $x = 5$ but only three ways to win instead of four, and it comes out just behind.

So we advise B to guess $\boxed{x = 5}$, with expected gain 437.5 pence, i.e. £4.375 (about £4.38) per game.

Checks

- The probabilities in the table sum to $\frac{16}{16} = 1$.
- Part (i) comes out exactly at 25, which confirms the table and formula.
- A simulation of 400 000 games guessing 5 gave an average gain of about 437.2 pence, against the exact 437.5.

Common slips

- Treating the totals 2 to 8 as equally likely: they are not, because the 16 *ordered* pairs are the equally likely outcomes.
- Forgetting that B loses $50x$ with probability $1 - p$, not with probability 1.
- Stopping at $x = 6$ or $x = 7$ because the prize is bigger: 5 and 6 are close, so every value must be computed.

Beyond STEP

This is a small decision problem: choose the action that maximises expected payoff. At university the same idea appears in decision theory and in the study of sums of independent random variables, where the distribution of a sum is found by *convolution* (here, the triangular shape of the table) and, for many dice, approaches a normal curve by the central limit theorem. Grimmett and Welsh, *Probability: An Introduction*, covers both.

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